



Time = 20 Minutes

Quiz No 01

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

**Question 1 [20 points] Sample Space and Probability Calculations**

**CLO Assessed – CLO 01, Transform a description of random experiment into formal setting of probability theory – construct sample space, identify and specify events. (Cog-4)**

Suppose you roll two fair six-sided die – one is a red die and the other is a green die. Both dice have numbers one to six. We define the following two random variables  $X$  &  $Y$ :

$X$  = number shown on the fair six-sided red die.

$$Y = \begin{cases} 0 & \text{if the two dice have one odd number and one even number.} \\ 1 & \text{if the two dice both have odd numbers.} \\ 2 & \text{if the two dice both have even numbers.} \end{cases}$$

- (a) [10 points] Capture (using any feasible means) the sample space showing all the possible outcomes  $(X, Y)$  and the joint probabilities for all valid values of  $(X, Y)$ .
- (b) [5 points] Let  $A$  be the event that the two dice both have even numbers. Calculate the  $P(A)$ .
- (c) [5 points] Let  $B$  be the event that the two dice have one odd number and one even number. Calculate the  $P(B)$ .

(a)

Green Die

Value of  $(x, y)$

Each box w. probability =  $1/36$

for working!

|   |     | 1       | 2   | 3   | 4   | 5   | 6 |
|---|-----|---------|-----|-----|-----|-----|---|
| 6 | 1,0 | 2,2     | 3,0 | 4,2 | 5,0 | 6,2 |   |
| 5 | 1,1 | 2,0     | 3,1 | 4,0 | 5,1 | 6,0 |   |
| 4 | 1,0 | 2,2     | 3,0 | 4,2 | 5,0 | 6,2 |   |
| 3 | 1,1 | 2,0     | 3,1 | 4,0 | 5,1 | 6,0 |   |
| 2 | 1,0 | 2,2     | 3,0 | 4,2 | 5,0 | 6,2 |   |
| 1 | 1,1 | 2,0     | 3,1 | 4,0 | 5,1 | 6,0 |   |
|   |     | 1       | 2   | 3   | 4   | 5   | 6 |
|   |     | Red Die |     |     |     |     |   |

Grid. 4pts

| Write $P_{X,Y}(x, y)$ |   | $X =$          |                |                |                |                |                |
|-----------------------|---|----------------|----------------|----------------|----------------|----------------|----------------|
|                       |   | 1              | 2              | 3              | 4              | 5              | 6              |
| $Y =$                 | 0 | $\frac{3}{36}$ | $\frac{3}{36}$ | $\frac{3}{36}$ | $\frac{3}{36}$ | $\frac{3}{36}$ | $\frac{3}{36}$ |
|                       | 1 | $\frac{3}{36}$ | 0              | $\frac{3}{36}$ | 0              | $\frac{3}{36}$ | 0              |
|                       | 2 | 0              | $\frac{3}{36}$ | 0              | $\frac{3}{36}$ | 0              | $\frac{3}{36}$ |

values 6pts

$$(b) P(A) = P(Y=2)$$

$$= 3 \cdot \frac{3}{36} = \frac{9}{36}$$

$$\overset{R}{(2,2)}, \overset{G}{(2,4)}, \overset{R}{(2,6)}, \overset{G}{(4,2)}, \overset{G}{(4,4)}, \overset{R}{(4,6)}, \overset{G}{(6,2)}, \overset{G}{(6,4)}, \overset{R}{(6,6)}$$

$$9 \times \frac{1}{36} = \frac{9}{36} = \frac{1}{4}$$

$$(c) P(B) = P(Y=0)$$

$$= 6 \cdot \frac{3}{36} = \frac{18}{36} = \frac{1}{2}$$

Red Dice  $\rightarrow 1, 2, 3, 4, 5, 6$

Green Dice  $\rightarrow (2,4,6) (1,3,5) \quad (6,1) (6,3), (6,5)$

$$6 \times 3 \times \frac{1}{36} = \frac{18}{36} = \frac{1}{2}$$